Prakash Musmade

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Education

GH RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT, PUNE *2023 – 2027*
Bachelor of Technology in Information Technology

EXPERIENCE

Full Stack Web Developer Intern · The EduNest, Pune *Jul 2025 – Present*
- •Developing responsive web interfaces and working on backend services and APIs using HTML5, CSS3, JavaScript, Node.js/Express, or PHP.
- •Working with MySQL databases and RESTful APIs while using Git/GitHub for version control and collaborative development.
- •Supporting testing, debugging, deployment, and environment configuration while collaborating with designers, developers, and project stakeholders.

Projects

IoT-Based Barcode Attendance System | ESP32 · C++ · FreeRTOS · Google Sheets API · Linear Regression · Chart.js
Apr 2026
GitHub
- •Architected a serverless IoT attendance system on a ₹435 ESP32 stack, replacing ₹5,000+ biometric hardware, supporting 50 concurrent students over local Wi-Fi.
- •Engineered dual-core FreeRTOS firmware: Core 1 serves 6 HTTP REST endpoints (<50 ms response); Core 0 asynchronously syncs attendance to Google Sheets via HTTPS with 3× retry logic.
- •Built a LittleFS-served real-time dashboard with 2-second auto-refresh, student search, session lock/unlock, and 3-format CSV export (daily / semester / below-35% alert list).
- •Implemented browser-side OLS Linear Regression on 8-week attendance histories to predict end-semester percentage; achieved 3.7 pp MAE and 85% risk classification accuracy across 20 students.
- •Validated over 4-week deployment with 60 students: 40-student class marked in <3 min vs. 8 – 12 min manual, dashboard latency 1.8 s avg — all 6 engineering targets met.

Daily Expense Tracker | HTML · CSS · PHP · MySQL · XAMPP
GitHub
- •Developed a full-stack web application for personal expense management with secure user authentication (login/signup) and CRUD operations for daily spending records.
- •Implemented server-side PHP logic with parameterised MySQL queries to prevent SQL injection; designed a normalised relational schema supporting multi-category filtering and date-range queries.

Skills

Languages: C++, JavaScript (ES6+), Java, SQL, PHP, Python
Frameworks / Libraries: HTML5, CSS3, React.js, Tailwind CSS, Chart.js, Fetch API, Axios, Google Apps Script
IoT & Embedded: ESP32, FreeRTOS, Arduino IDE 2.x, LittleFS, GPIO, UART, I2C, SPI
Databases: MySQL, Google Sheets (cloud)
Tools: Git, VS Code, Arduino IDE, Chrome DevTools, Docker, XAMPP, Postman
Coursework: Data Structures & Algorithms, OOP, Computer Networks, DBMS, Machine Learning, IoT Systems, Web Technologies